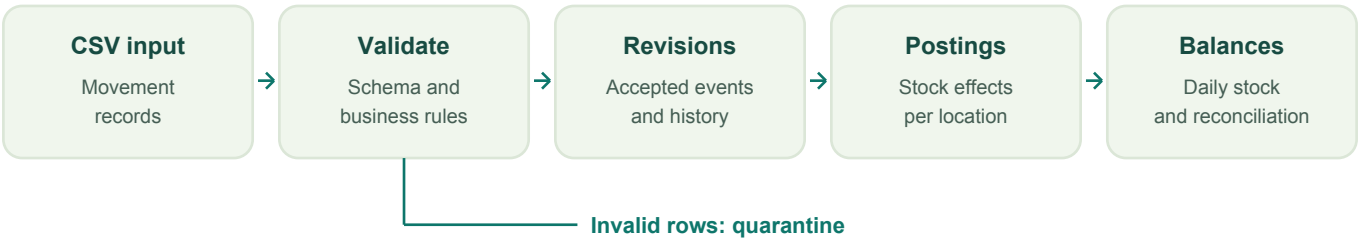
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Understand Stock Invento

A practical guide to the screens, stock calculations, import exercises and code.

Start with one question: what happened to an item at a location, and can we explain the resulting stock? Stock Invento turns movement CSVs into daily balances while preserving corrections, rejected rows and an operational history.

Follow one import through the system



Every import attempt also records its counts, outcome, source and timing.

The calculation behind every daily row

```
Closing = Opening + Receipts - Issues
+ Transfers in - Transfers out + Adjustments
```

The next day starts with the previous day's closing. A daily row belongs to exactly one **item, location and business date**.

Movement	Effect on stock
Receipt	Adds stock at the receiving location.
Issue	Removes stock from the issuing location.
Transfer	Removes from the source and adds equally to the destination.
Adjustment	Adds or subtracts the signed quantity.

Exact quantities: the app accepts up to three decimal places and calculates in integer thousandths. For example, 12.345 is stored as 12345. A transfer preserves total stock for that item across its two locations.

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Read the dashboard

Open the local app and sign in. The main header's **Import CSV** button opens the drawer. Choosing a file only previews it; **Import & publish** submits it. Close or Escape dismisses the drawer, and reopening keeps its draft while the page stays open.

50
CURRENT EVENTS
Latest non-void movements

53
HISTORICAL VERSIONS
Originals plus revisions

47
DAILY BALANCE ROWS
Item + location + day

These are fresh seeded-demo counts, not interchangeable totals. They change after your own imports. Corrections add versions; daily balance rows are derived results.

View	What to inspect
Daily balances	Stock by item, location and day. Filter PRACTICE-ITEM during the exercises. Export view downloads the filtered balance view.
Correction impact	Before/after balances caused by a selected import. Later days can change when an earlier event is corrected.
Reconciliation	Whether accepted movements, stock postings and daily equations agree.
Pipeline runs	Each import attempt: status, source, timing, accepted, duplicate, rejected and rolled-back counts.
Quarantine	Rejected records, original values, source line and validation reason.
Event history	Original, superseded, voided, restored and current movement revisions.

Keep these distinctions clear

Negative stock is a warning. It can result from valid accepted movements. A quarantined row is invalid and contributes no stock.

Passing reconciliation proves consistency. It does not mean every input was accepted or that stock availability is operationally acceptable.

Review required is dynamic and includes history. Existing rejects and failed runs remain visible after a later success. The seeded demo intentionally includes learning examples. Process guide and Achievements & verification are collapsed disclosures below the ledger.

Learn with the included CSVs

Start a separate learning workspace from the extracted folder containing run_demo.py. Use an unused data-folder name.

```
python3 run_demo.py --data-dir var/learning-new --port 8766
```

Open http://127.0.0.1:8766/login. Sign in with **admin / admin123**. On Windows, use **py -3** instead of python3. Keep the terminal open.

Load catalogs first. In Import CSV, expand **Using new item or location IDs?** Select `00_items.csv` and `00_locations.csv` in their catalog pickers, then Validate & load catalogs. Files are in `data/import-practice/`.

Next import movement files one at a time in this order. Leave Publish through date blank; the fresh demo retains its January 4 horizon. Filter **PRACTICE-ITEM** on **2026-01-04**. MAIN and BRANCH below mean PRACTICE-MAIN and PRACTICE-BRANCH.

Movement file	First-import outcome	MAIN	BRANCH
`01_receipts.csv`	2 accepted Receipts	100.000	25.000
`02_issues_transfers_adjustments.csv`	3 accepted Issue, transfer, adjustment	62.500	45.000
`03_backdated_correction.csv`	1 accepted Receipt revision 2	82.500	45.000
`04_mixed_valid_and_invalid.csv`	1 accepted 3 quarantined	82.500	55.000
`05_negative_stock.csv`	1 accepted Negative stock	-17.500	55.000
`06_fix_negative_stock.csv`	1 accepted Issue revision 2	22.500	55.000

```
After file 03:
MAIN = 120 - 15 - 20 - 2.500 = 82.500
BRANCH = 25 + 20 = 45.000
```

Trace, replay, then inspect the edge cases

After each import, follow **Pipeline runs - Event history - Daily balances - Correction impact - Reconciliation**. Reimport a valid file: accepted 0, valid rows counted as duplicates, unchanged stock. A new audit attempt remains.

Files 04-06 are optional. File 04 rejects an unknown item, excess precision and a self-transfer. File 05 creates negative stock; file 06 corrects it. Replaying file 04 quarantines its three invalid rows again. Seeded warnings can remain after the practice negative stock is fixed.

Answer sheet - do not upload: data/expected/import_practice_balances.csv.
Full steps: data/import-practice/README.md. Practice IDs are separate from the seeded demo.

Connect the behavior to the code

The interface is HTML/CSS/JavaScript. A local Python HTTP server calls the processing pipeline, and SQLite stores references, versions, audit records, postings and balances. The app uses the Python standard library; no external account is required.

Read in this order	Responsibility
`inventory/cli.py`	Commands for setup, demo, import, reports, export and serving.
`inventory/model.py`	CSV fields, identifiers, references, dates and exact quantities.
`inventory/pipeline.py`	Ingestion, immutable revisions, transactions, balance rebuilds and reconciliation.
`inventory/impact.py`	Retained before/after comparisons for each successful publication.
`inventory/web.py`	Local HTTP routes, API validation and browser import requests.
`inventory/auth.py`	Password hashes, sign-in, sessions, expiry and logout.
`inventory/static/app.js`	Dashboard rendering, filters, drawer, previews and API requests.

Explain these four rules in your own words

1. Revision: a correction is a complete replacement with a higher revision. The highest accepted revision controls the current event; previous versions remain. A current voided event contributes no stock. Changing an already stored revision's payload creates a conflict.

2. Replay: identical event/revision pairs are duplicates, so stock is not multiplied. Import attempts and repeated rejected rows still add audit history.

3. Failure: valid rows in a mixed file can publish together while invalid rows are quarantined. A fatal batch error rolls back its inventory publication and preserves the prior ledger, while failure evidence remains.

4. Trade-off: publication currently rebuilds daily balances. Updating only affected item/location/date portions is a possible future optimization, not an implemented feature.

Finish your learning walkthrough

Demonstrate a correction changing later days, a replay changing no stock, and negative stock coexisting with passing reconciliation. Then run the practice verifier in a temporary workspace:

```
python3 scripts/verify_practice_imports.py
```

Read next: **START_HERE.txt**, **docs/walkthrough.md**, **docs/data-dictionary.md**, and **docs/design.md**. The ZIP includes test scenarios and reports.

Local scope: login uses a password hash and server-side sessions, not OAuth. It protects HTTP access, not direct OS file/CLI access. The SQLite database is not encrypted by this login. Runtime verification was on macOS.